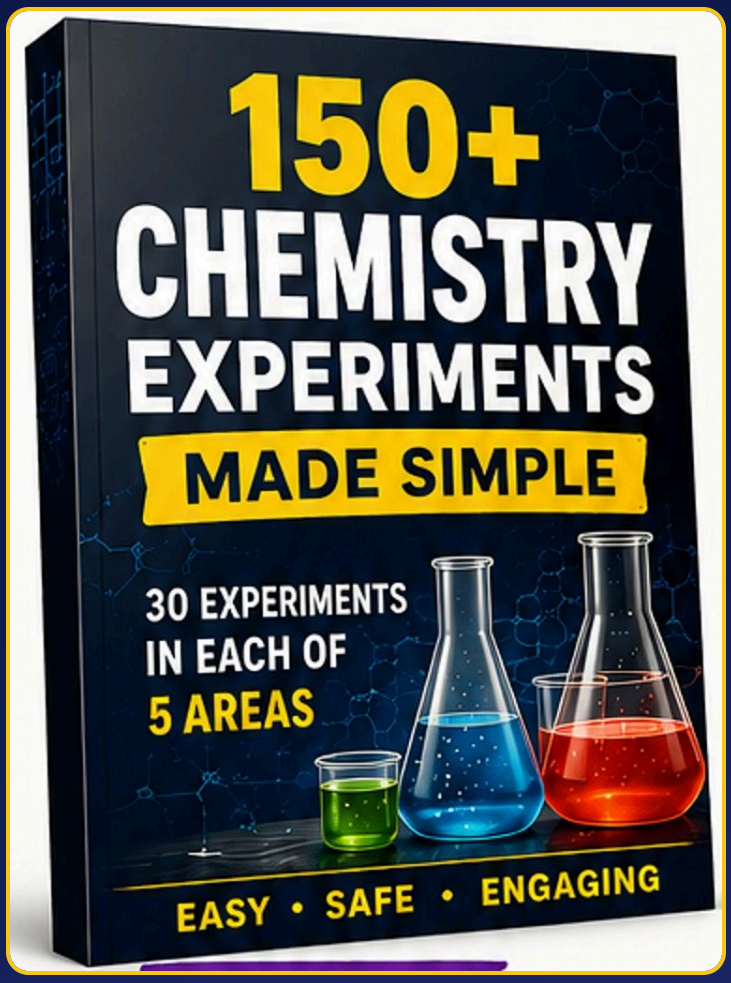
BONUS 5

Assessment Sheets + Rubrics for Hands-On Labs

Observation sheets, CER template, 4-point rubric, group checklist and exit tickets



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How to Use Hands-On Assessment

> *A hands-on lab produces more evidence of learning than almost any worksheet, but only if you collect that evidence on purpose. This bonus gives you ready-to-print sheets, rubrics, and mini-assessments that fit the one-page experiment cards, so you can assess without adding a heavy grading load.*

Formative vs. summative: a quick overview

	FORMATIVE ASSESSMENT	SUMMATIVE ASSESSMENT
Purpose	Find out what students understand right now so you can adjust teaching	Judge what students have learned at the end of a unit or project
When	During the lab or at the end of the class period	After several labs, at the end of a unit, or for a lab report
Typical tools in this bonus	Observation sheets, exit tickets, peer review, self-assessment, quick feedback stems	4-point rubrics, lab report rubric, CER writing, data-literacy quiz
Stakes	Low. Often not graded, or graded for completion	Higher. Converted to a grade with the conversion table
Teacher time	1 to 3 minutes per class	10 to 20 minutes per student product

A good rhythm for one experiment card is simple. Use an **observation sheet** while students work, a **CER sheet** or **exit ticket** at the end of class, and a **rubric** only when a product is going to count toward a grade. Most labs need just two of these tools, not all of them.

A simple weekly routine

1. **Before the lab:** choose one skill to watch (for example, measuring or recording). Tell students what it is.
2. **During the lab:** walk the room with the observation log and mark only that skill. Aim to see every student once every two or three labs.
3. **After the lab:** students complete a CER sheet or an exit ticket in the last five minutes.
4. **End of unit:** score one product (a lab report or a CER paragraph) with the matching rubric and convert it to a grade.

> **Assess the skill, not the result**
>
> Experiments sometimes do not give the expected result. A student who measures carefully, records honestly, and explains an unexpected result well has shown more science than a student who copies the textbook answer. The rubrics in this bonus reward process and reasoning.

> **Safety is always assessed**
>
> Every rubric set includes safety and lab behavior. Students should know from the first lab that safe practice is part of the grade, not an extra.

Observation Sheets

Three versions follow. Print the one that matches your class. Each sheet works with any experiment card, and each can be copied as many times as you need.

Version A: Student Observation Sheet, Grades 6-8 (simple)

Observation Sheet (Grades 6-8)

Name: _____ Date: _____

Experiment title / card number: _____

My question: What do I want to find out?

My prediction: I think _____ will happen because _____

Materials I used: _____

Before (what I saw, felt, or smelled before mixing; never taste): _____

During (what changed while we worked): _____

After (what it looked like at the end): _____

Did anything surprise me? YES NO What? _____

Draw a labeled picture of the result in the box on the back of this page.

What did I learn? One sentence: _____

_____

_____

_____

_____

_____

_____

Version B: Student Observation Sheet, Grades 9-12 (detailed)

Observation and Data Sheet (Grades 9-12)

Name: _____ Date: _____ Class period: _____

Lab title and card ID: _____ Partners: _____

Research question: _____

Hypothesis (If ... then ... because ...): _____

Independent variable: _____ Dependent variable: _____

Controlled variables (list at least three): _____

Equipment and amounts (include units): _____

Qualitative observations (color, clarity, bubbles, smell by wafting only, texture): _____

Quantitative data (include units and uncertainty, for example +/- 0.5 mL): _____

Trial 1: _____ Trial 2: _____ Trial 3: _____ Average: _____

Sources of error (at least two) and how to reduce them: _____

Evidence of a chemical change? Name the signs you saw: _____

Conclusion in 2-3 sentences, using your data: _____

_____

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Version C: Teacher Observation Log (class of 30)

Print one page per lab. Write the date and card ID at the top. Use the code **3** = independent, **2** = with a reminder, **1** = needs help, and leave the box blank if you did not observe that student. Add a short note only when it matters. Copy the table onto a landscape page for easier use.

# STUDENT	FOLLOWS SAFETY STEPS	MEASURES ACCURATELY	OBSERVES AND RECORDS	WORKS WITH PARTNER	USES SCIENCE VOCABULARY	NOTE
1						
2						
3						
4						
5						

# STUDENT	FOLLOWS SAFETY STEPS	MEASURES ACCURATELY	OBSERVES AND RECORDS	WORKS WITH PARTNER	USES SCIENCE VOCABULARY	NOTE
6						
7						
8						
9						
10						
11						
12						
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27						
28						
29						
30						

> **Do not try to score everyone every day**
>
> Pick six to eight students per lab. In one week of three labs you will have seen the whole class at least once. Watch for patterns, such as a student who is always the recorder or never touches materials, and fix those through role rotation.

CER Template: Claim, Evidence, Reasoning

CER gives students a short, repeatable structure for explaining results. The **claim** answers the question. The **evidence** is the data or observations that support it. The **reasoning** connects the evidence to a science idea and explains why the evidence supports the claim.

CER Student Sheet

Name: _____ Date: _____

Question we are answering: _____

CLAIM (one sentence that answers the question): _____

EVIDENCE (at least two pieces of data or observations; include numbers and units): _____

REASONING (which science idea links your evidence to your claim, and why?): _____

Counterclaim check: What could someone argue against your claim, and how would you respond? _____

_____

_____

_____

_____

_____

_____

_____

Worked example: Does mass change when baking soda and vinegar react in an open container?

Setup: Students place 1 tbsp (about 14 g) of baking soda in one cup and 1/2 cup (120 mL) of white vinegar in a second cup, both sitting on a scale. The total mass is recorded. They then pour the vinegar into the baking soda, let the fizzing finish, and read the scale again. Use an open container only, with goggles on and the work done on a tray.

MEASUREMENT	MASS (G)
Both cups with contents, before mixing	312.0
Both cups with contents, after fizzing stops	309.6
Change in mass	-2.4

Model CER response

Claim: The total mass went down because a gas that formed left the open container, not because matter disappeared.
Evidence: The scale read 312.0 g before mixing and 309.6 g after the fizzing stopped, a loss of 2.4 g. We saw many bubbles rising out of the cup while it fizzed.
Reasoning: Baking soda and vinegar react to make carbon dioxide gas, water, and other products. In an open container the gas escapes into the air, and the scale cannot weigh it. The law of conservation of mass says the total mass of the atoms stays the same in a reaction, so the missing 2.4 g must be the mass of the gas that left.

Second worked example: Which liquid is denser?

Setup: Students pour equal amounts of water, vegetable oil, and dish soap into a clear jar and watch how they layer. They also use a scale and a measuring cup to find the mass of 1/4 cup (60 mL) of each liquid.

LIQUID	VOLUME (ML)	MASS (G)	DENSITY (G/ML)
Water	60	60	1.00
Vegetable oil	60	55	0.92
Dish soap	60	64	1.07

Model CER response

Claim: Dish soap is denser than water, and vegetable oil is less dense than water.
Evidence: The same 60 mL of dish soap had a mass of 64 g (1.07 g/mL), water had 60 g (1.00 g/mL), and oil had 55 g (0.92 g/mL). In the jar, the oil floated on top of the water and the soap sank to the bottom.
Reasoning: Density is mass divided by volume. A liquid with more mass in the same volume has more matter packed into that space, so it sinks below a liquid with a lower density. The layering order matches the order of the calculated densities.

Sentence-starter bank

PART	SENTENCE STARTERS
Claim	I claim that ... | The answer to our question is ... | Our results show that ...
Evidence	The data show ... | When we measured ..., we found ... | We observed ... which means ... | In trial 2, the number changed from ... to ...
Reasoning	This evidence supports my claim because ... | The science idea that explains this is ... | This happens because the particles ... | This matches the idea that ...
Counterclaim	Someone might say ..., but the data show ... | An error that could affect this is ..., however ...
Uncertainty	I am fairly sure because ... | To be more certain we would need ...

4-Point Rubrics

Each rubric uses the same four levels so students learn one language of quality: **4 = Exceeds**, **3 = Meets**, **2 = Approaching**, **1 = Beginning**. Share the rubric before the lab, not after. For quick scoring, circle the level that best fits and add one comment.

Rubric 1: Lab Investigation

CRITERION	4 EXCEEDS	3 MEETS	2 APPROACHING	1 BEGINNING
Question and prediction	Testable question with a prediction explained by a science idea	Testable question with a clear prediction	Question or prediction is vague or not testable	Missing or unrelated to the lab
Procedure	Follows every step accurately and improves the method with a sensible adjustment	Follows all steps in order with accurate measurements	Skips or reorders a step; some measurements are rough	Needs repeated help to complete the procedure
Observations	Detailed, accurate, and includes both numbers and descriptions	Accurate and complete with some detail	Some observations are missing or general	Few or no observations recorded
Conclusion	Answers the question using data and addresses an error or new question	Answers the question using data	Partial answer; data not clearly used	No answer or not based on the lab

Rubric 2: CER Writing

CRITERION	4 EXCEEDS	3 MEETS	2 APPROACHING	1 BEGINNING
Claim	Precise claim that fully answers the question	Clear claim that answers the question	Claim is partly correct or incomplete	No claim or it does not match the question
Evidence	Three or more accurate pieces of evidence with units	Two accurate pieces of evidence	One piece of evidence, or evidence is vague	No evidence, or it does not support the claim
Reasoning	Uses a correct science idea and explains the link in detail	Uses a correct science idea to link evidence and claim	Mentions an idea but does not connect it	No reasoning, or reasoning is incorrect
Writing	Organized, uses vocabulary correctly throughout	Organized with correct key vocabulary	Some vocabulary missing or misused	Hard to follow

Rubric 3: Data Table and Graph

CRITERION	4 EXCEEDS	3 MEETS	2 APPROACHING	1 BEGINNING
Data table	Titled, labeled with units, organized, and includes averages	Labeled with units and organized	Missing a unit or label; hard to read	Data are scattered or missing
Accuracy	All values match the observations	Most values match; minor slips	Several errors or missing values	Most values are inaccurate
Graph	Correct type, titled, axes labeled with scale and units, points accurate	Correct type, labeled axes, mostly accurate	Missing a label or scale problem	Graph missing or unreadable
Interpretation	Describes the trend and uses numbers to support it	Describes the trend	Mentions the data without a trend	No interpretation

Rubric 4: Scientific Explanation

CRITERION	4 EXCEEDS	3 MEETS	2 APPROACHING	1 BEGINNING
Science ideas	Accurate and connected to particles, energy, or atoms	Accurate main idea	Some correct ideas with errors	Mostly incorrect or missing
Vocabulary	Uses terms precisely and defines them in own words	Uses terms correctly	Uses some terms incorrectly	Little or no science vocabulary
Connections	Links the lab to an everyday example	Links cause and effect clearly	Cause and effect is unclear	No cause and effect
Clarity	Clear, logical, and concise	Clear and logical	Parts are confusing	Not understandable

Rubric 5: Safety and Lab Behavior

CRITERION	4 EXCEEDS	3 MEETS	2 APPROACHING	1 BEGINNING
Protective gear	Wears goggles correctly at all times and reminds others	Wears goggles correctly at all times	Needs one reminder	Needs repeated reminders
Following instructions	Follows all safety instructions without prompting	Follows safety instructions	Follows most; needs reminders	Ignores safety instructions
Materials handling	Handles, pours, and cleans up carefully; reports spills at once	Handles materials carefully and cleans up	Some spills or messy cleanup	Unsafe handling of materials
Teamwork and focus	Focused and helps keep the group safe	Focused on the task	Sometimes off task	Distracts others from safe work

Rubric 6: Lab Report

CRITERION	4 EXCEEDS	3 MEETS	2 APPROACHING	1 BEGINNING
Title and purpose	Clear title and a focused purpose statement with a testable question	Title and clear purpose	Purpose is incomplete	Missing
Materials and procedure	Complete list with amounts and a procedure someone else could repeat	Complete list and a clear procedure	Some amounts or steps missing	Missing or incorrect
Data and analysis	Organized table, graph, and calculations with units; discusses error	Table and graph with units and a clear analysis	Data present but analysis incomplete	Little or no data
Conclusion (CER)	Strong claim, evidence, and reasoning; suggests next steps	Claim, evidence, and reasoning present	One part of CER missing	No conclusion
Presentation	Neat, organized, and carefully proofread	Neat and organized	Hard to read in places	Disorganized

Group Work Checklist

Many experiments run best in groups of four or five, and every student should have a role. Rotate roles every lab so everyone practices every job.

ROLE	MAIN JOB
Manager	Reads each step aloud, keeps the group on time, and is the only person who asks the teacher a question
Materials	Collects and returns materials on a tray, measures with the group, and keeps the area tidy
Recorder	Writes data, observations, and drawings on the sheet and reads them back to confirm
Speaker	Shares the group results and explanation with the class
Safety officer	Checks goggles, watches for spills, reminds the group of the safety note, and signals the teacher if help is needed

STUDENT GROUP CHECKLIST (CHECK EACH ITEM WHEN DONE)

☐ Everyone is wearing splash goggles and long hair is tied back.

☐ Roles are assigned and written on the sheet.

☐ We read the whole card, including the safety note, before starting.

☐ All materials are on the tray and the work area is clear.

☐ We predicted what will happen before starting.

☐ Everyone had a turn doing a hands-on task.

☐ We recorded data and observations during the lab.

☐ We cleaned up, washed hands, and returned materials.

☐ We agreed on a CER answer before leaving.

TEACHER CHECKLIST FOR GROUP WORK

☐ Groups are of four or five, mixed by skill level.

☐ Roles are posted on the board and rotated each lab.

☐ Safety rules are reviewed out loud before materials are handed out.

☐ Materials are pre-measured or placed at a station to save time.

☐ I circulate and check each group at least once.

☐ I have a signal (hand up or timer) for stopping and cleaning.

☐ I used the observation log for at least six students.

☐ Every group shared a result before the end of class.

Peer Review Form

Peer Review of a Lab Write-Up

Reviewer name: _____ Author name: _____ Date: _____

Does the claim answer the question? YES SOMEWHAT NO

Two pieces of evidence that I found: 1. _____ 2. _____

Does the reasoning explain WHY the evidence supports the claim? YES SOMEWHAT NO

One thing that is done well (be specific): _____

One question I still have: _____

One suggestion to make it stronger: _____

Are the units and labels correct in the data table? YES NO Where? _____

_____

_____

_____

_____

_____

Set the tone

Review the work, not the person. Use the stems: I noticed ..., I wonder ..., A next step could be ... Practice one review aloud as a class before students swap papers.

Self-Assessment Form

My Lab Self-Check

Name: _____ Date: _____ Lab: _____

Circle one for each: 4 = Very confident, 3 = Confident, 2 = A little unsure, 1 = Need help

I wore my goggles and followed safety steps. 4 3 2 1

I measured carefully and used the right units. 4 3 2 1

I recorded what I saw as it happened. 4 3 2 1

I can explain what happened using science words. 4 3 2 1

I worked well with my group. 4 3 2 1

The part I am proudest of: _____

The part I want to get better at: _____

One question I still have about this experiment: _____

Exit Tickets: 20 Ready-to-Copy Mini-Assessments

Each exit ticket takes about three minutes. Print them and hand one out in the last five minutes of class, then collect it at the door. Sort the tickets into three piles: **got it**, **almost**, and **needs reteaching**. Sample answers are in the key at the end of this section.

Exit Ticket 1: States of Matter

Name: _____ Date: _____

1. Name the three common states of matter and tell how the particles move in a solid.

2. What happens to particles when a solid is heated until it melts?

Exit Ticket 2: Density

Name: _____ Date: _____

1. Two liquids are in a jar. Liquid A floats on liquid B. Which has the greater density?

2. A block has a mass of 24 g and a volume of 8 mL. What is its density?

Exit Ticket 3: pH

Name: _____ Date: _____

1. Is a solution with a pH of 3 acidic, basic, or neutral?

2. Name one household item that is usually basic.

Exit Ticket 4: Signs of a Reaction

Name: _____ Date: _____

1. List two signs that a chemical reaction may have happened.

2. A solid dissolves in water and no new substance forms. Is that a chemical reaction? Explain.

Exit Ticket 5: Conservation of Mass

Name: _____ Date: _____

1. In a sealed container, the total mass before and after a reaction is 50 g. What does that tell you about the atoms?

2. In an open container the mass seems to drop. Give a reason.

Exit Ticket 6: Catalysts

Name: _____ Date: _____

1. What does a catalyst do to a reaction?
2. Is a catalyst used up in the reaction? Explain.

Exit Ticket 7: Reaction Rate

Name: _____ Date: _____

1. Name two things that can speed up a reaction.
2. Why does a powdered solid dissolve faster than one large chunk?

Exit Ticket 8: Solubility

Name: _____ Date: _____

1. What does it mean when a substance is soluble?
2. How does water temperature usually change how much sugar can dissolve?

Exit Ticket 9: Mixtures vs. Solutions

Name: _____ Date: _____

1. Is salt water a mixture or a solution? Explain.
2. Name one way to separate a mixture of sand and water.

Exit Ticket 10: Physical vs. Chemical Change

Name: _____ Date: _____

1. Is tearing paper a physical or a chemical change?
2. Is baking a cake a physical or a chemical change? Give one clue.

Exit Ticket 11: Indicators

Name: _____ Date: _____

1. What does an indicator do?
2. Red cabbage juice turns pink in a clear liquid. What can you infer about the liquid?

Exit Ticket 12: Endothermic and Exothermic

Name: _____ Date: _____

1. Which process feels cold to the touch: endothermic or exothermic?
2. A reaction gives off heat to the surroundings. Is it exothermic or endothermic?

Exit Ticket 13: Polymers

Name: _____ Date: _____

1. What is a polymer?
2. Name one natural polymer and one synthetic polymer.

Exit Ticket 14: Diffusion

Name: _____ Date: _____

1. Define diffusion in your own words.

2. Will food coloring spread faster in hot or cold water? Why?

Exit Ticket 15: Osmosis

Name: _____ Date: _____

1. What substance moves across the membrane in osmosis?

2. A potato slice is left in very salty water. What do you predict happens to it?

Exit Ticket 16: Evaporation

Name: _____ Date: _____

1. What is evaporation?

2. Name two things that make a puddle evaporate faster.

Exit Ticket 17: Crystals

Name: _____ Date: _____

1. Explain how crystals form from a saturated solution as it cools.

2. Why do slow-growing crystals often look larger and clearer?

Exit Ticket 18: Freezing Point Depression

Name: _____ Date: _____

1. When salt is added to ice, does the melting temperature go up or down?

2. Why is salt spread on icy roads?

Exit Ticket 19: Metals and Oxidation

Name: _____ Date: _____

1. What is rust, and what two things must be present for iron to rust?

2. Name one way to slow rusting.

Exit Ticket 20: Water Filtering

Name: _____ Date: _____

1. Name two materials that can filter muddy water.

2. Does filtering make water safe to drink? Explain.

Exit Ticket Answer Key

TICKET	SAMPLE ANSWERS (Q1 | Q2)
1. States of Matter	Solid, liquid, gas. In a solid the particles vibrate in place. | The particles gain energy, vibrate faster, and break free of fixed positions to slide past each other.
2. Density	Liquid B is denser. | 24 / 8 = 3 g/mL.
3. pH	Acidic. | Baking soda solution (also soap or an antacid).
4. Signs of a Reaction	Any two: gas bubbles, color change, temperature change, new solid forms, light. | No. It is a physical change: the substance spreads out but is still the same substance.

TICKET	SAMPLE ANSWERS (Q1 | Q2)
5. Conservation of Mass	The total mass of the atoms stays the same; atoms are only rearranged. | A gas escaped from the container.
6. Catalysts	It speeds up a reaction without being used up. | No. It is still there at the end and can work again.
7. Reaction Rate	Any two: higher temperature, higher concentration, smaller pieces, a catalyst. | Powder has more surface area, so more particles collide with the water.
8. Solubility	It can dissolve in a liquid to form a solution. | Warmer water usually dissolves more sugar.
9. Mixtures vs. Solutions	A solution, because it is evenly mixed at the particle level. | Filtering, or evaporating the water.
10. Physical vs. Chemical Change	Physical. | Chemical; clues: new substance, color and smell change, cannot be reversed.
11. Indicators	It changes color to show whether something is an acid or a base. | The liquid is acidic.
12. Endothermic and Exothermic	Endothermic. | Exothermic.
13. Polymers	A large molecule made of many repeating units. | Natural: starch, cellulose, or rubber. Synthetic: plastic or nylon.
14. Diffusion	The spreading of particles from high to low concentration. | Hot, because particles move faster.
15. Osmosis	Water. | It shrinks and becomes soft because water leaves the cells.
16. Evaporation	Liquid water changing to gas below the boiling point. | Any two: warmer temperature, wind, more surface area, drier air.
17. Crystals	As it cools, less solute can stay dissolved, so particles join an orderly pattern. | Slow growth gives particles time to fit into the pattern neatly.
18. Freezing Point Depression	Down. | It lowers the freezing point, so the ice melts at a lower temperature.
19. Metals and Oxidation	Iron oxide; iron needs oxygen and water. | Paint, oil, or dry storage.
20. Water Filtering	Sand, gravel, cloth, coffee filter, or cotton. | No. Filtering removes particles but may not remove germs or dissolved substances.

Data-Literacy Mini-Quiz

Data-Literacy Mini-Quiz (10 minutes)

Name: _____ Date: _____

Use this data: Plain water at 68 F (20 C) dissolved 1 tsp of sugar in 40 seconds. Water at 104 F (40 C) dissolved it in 22 seconds. Water at 140 F (60 C) dissolved it in 12 seconds. (An adult prepares the warm water.)

1. What is the independent variable? What is the dependent variable?
2. Which variables should stay the same to make this a fair test?
3. Describe the trend in the data using numbers.
4. Predict the time at 176 F (80 C). Is your prediction an interpolation or an extrapolation?
5. What is one way to make the data more trustworthy?

Quiz answer key

1. Independent: water temperature. Dependent: time to dissolve. 2. Amount and type of sugar, amount of water, stirring, and container. 3. As temperature rises from 68 F to 140 F, the time falls from 40 s to 12 s. 4. A time lower than 12 s (about 6 to 8 s); this is an extrapolation. 5. Repeat each temperature three times and average the results.

Rubric-to-Grade Conversion Table

Score each criterion from 1 to 4, add the points, and divide by the total possible. Use whichever method matches your gradebook.

AVERAGE RUBRIC SCORE	PERCENT	LETTER GRADE	MEANING
3.7 to 4.0	93 to 100%	A	Exceeds expectations
3.3 to 3.6	85 to 92%	B+ / A-	Meets expectations strongly
2.9 to 3.2	78 to 84%	B	Meets expectations
2.5 to 2.8	70 to 77%	C	Approaching
2.0 to 2.4	60 to 69%	D	Beginning

AVERAGE RUBRIC SCORE	PERCENT	LETTER GRADE	MEANING
Below 2.0	Below 60%	F	Not yet shown

Example: A lab report with five criteria is scored 4, 3, 3, 4, 3. The total is 17 of 20, or 85 percent, which falls in the B+ range. Many teachers also allow students to revise and resubmit, replacing the score with the new one.

How to Give Quick Feedback

The best feedback is short, specific, and aimed at the next step. Use these sentence stems on sticky notes, in the margin, or aloud as you circulate.

PURPOSE	SENTENCE STEMS
Praise the process	I noticed that you … | Your careful measuring of … helped … | I like how you checked your result by …
Push thinking	What evidence do you have for … ? | How do you know … ? | What would happen if … ?
Fix a mistake	Check the units on … | Look at your data in trial 2; does it fit your claim? | Reread step 3 of the card.
Connect to science	Which particle idea explains that? | How does this relate to conservation of mass? | Where else do you see this in daily life?
Set a next step	Next time, try … | Add one more piece of evidence about … | Revise your reasoning to include …

FEEDBACK TIPS

- ☐ Give one strength and one next step, not ten comments.
- ☐ Focus on the criterion from the rubric you shared before the lab.
- ☐ Give feedback while students can still use it.
- ☐ Ask students to respond to feedback with one change.

Next step

Choose one experiment card for this week. Print the observation sheet for your grade band, pick one rubric row to watch, and finish the class with an exit ticket. Review the tickets the same day and use the three piles to plan the first five minutes of the next class.